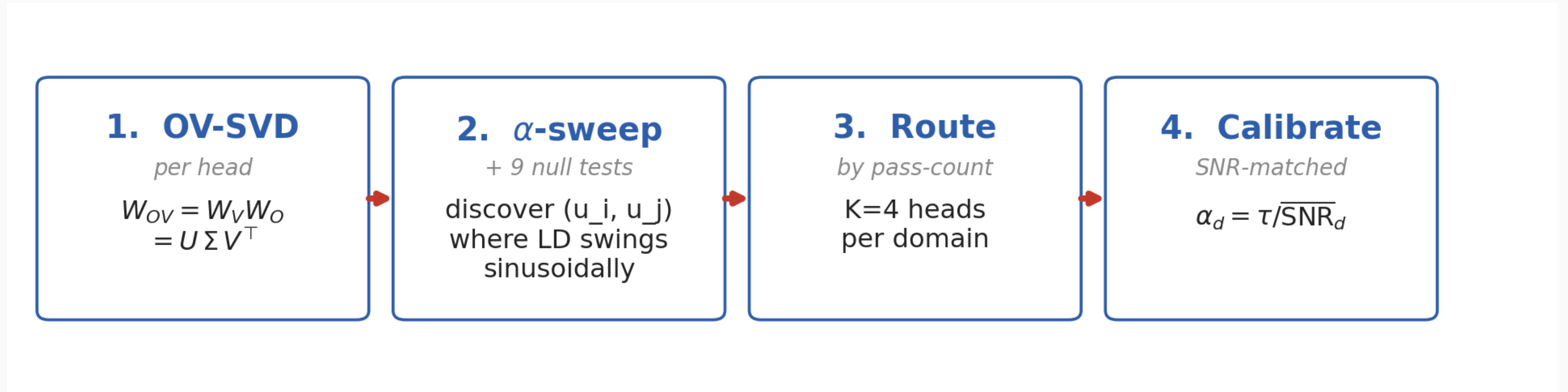


5. Pipeline

From raw weights to a routed debiasing dial



- **OV-SVD** per head → **α-sweep + 9 null tests** → **route by pass-count** → **calibrate** $\alpha_d = \tau / \overline{\text{SNR}}_d$.
- Blind-scan pass rates vs. full-OV null (95% CI): GPT-2 2.06%, Phi-3 1.18%, Gemma 2.02%, Llama 0.22% — all significantly above the null.
- Nine null tests knock out correlational planes: permuted labels, random planes (1000 MC), full-OV ablate, semantic-ablate, principal-angles, causal-patch, self-test, decode-heads, cyclicity.